

Q34. Banking Loan Approval Risk Optimization

1. Aim

To develop a Decision Tree-based loan approval system that evaluates applicant financial factors and classifies loan applications as **Approved** or **Rejected** while minimizing default risk.

2. Problem Statement

A bank receives a large number of loan applications and needs to decide whether each application should be approved or rejected. The decision should consider factors such as credit score, income, debt-to-income ratio, and previous loan defaults. An algorithmic decision-making system can help the bank make faster and more consistent decisions while reducing financial risk.

3. Objectives

- To analyze important loan applicant risk factors.
- To use a Decision Tree for loan approval decisions.
- To reduce the possibility of loan defaults.
- To support faster and consistent decision-making.
- To evaluate the scalability and computational complexity of the solution.

4. Constraints / Input Factors

The following factors are considered:

Factor	Description
Credit Score	Indicates the applicant's creditworthiness.

Income	Shows the applicant's repayment capacity.
Debt-to-Income Ratio	Indicates existing financial obligations.
Previous Default	Shows whether the applicant has defaulted on previous loans.

5. Algorithm Used

Decision Tree Classification

A **Decision Tree** is used to classify loan applications into two categories:

- **1** → **Loan Approved**
- **0** → **Loan Rejected**

The algorithm makes decisions by checking conditions on applicant features and follows branches until it reaches a final decision.

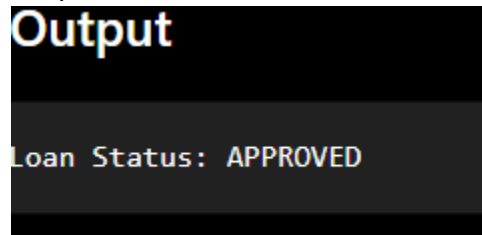
6. Algorithm

1. Start.
2. Collect loan applicant information.
3. Store credit score, income, debt-to-income ratio, and previous default information.
4. Prepare the training dataset.
5. Train a Decision Tree classification model.
6. Provide the new applicant's information to the trained model.
7. Predict the loan status.
8. If the prediction is **1**, display **Loan Approved**.
9. Otherwise, display **Loan Rejected**.
10. Stop.

Program:

```
1 from sklearn.tree import DecisionTreeClassifier
2 X = [
3     [750, 60000, 20, 0],
4     [680, 45000, 40, 0],
5     [620, 30000, 60, 1],
6     [800, 80000, 15, 0],
7     [580, 25000, 70, 1],
8     [720, 55000, 25, 0],
9     [650, 40000, 50, 1],
10    [780, 70000, 18, 0]
11 ]
12
13 # 1 = Approved, 0 = Rejected
14 y = [1, 1, 0, 1, 0, 1, 0, 1]
15
16 # Create Decision Tree
17 model = DecisionTreeClassifier(max_depth=3, random_state=42)
18
19 # Train the model
20 model.fit(X, y)
21
22 # New loan application
23 new_application = [[710, 50000, 30, 0]]
24
25 # Prediction
```

Output:



Output

Loan Status: APPROVED

9. Analysis

The Decision Tree analyzes the applicant's financial information and produces a loan approval decision. In the given example, the applicant has a good credit score, sufficient income, a relatively low debt-to-income ratio, and no previous default. Therefore, the model classifies the application as **Approved**.

10. Scalability

The system can be scaled for large numbers of loan applications by:

- Using efficient data preprocessing.
- Selecting only important features.
- Training the Decision Tree on large datasets.
- Using distributed computing when the dataset becomes very large.
- Periodically updating the model using new loan repayment data.

11. Computational Complexity

Let:

- n = number of loan applications
- m = number of features

The approximate training complexity of a balanced Decision Tree is:

$$O(n \times m \times \log n)$$

The prediction complexity is approximately:

$$O(\log n) \text{ for a balanced tree.}$$

The space requirement depends on the size of the generated tree.

12. Financial Risk Interpretation

The system helps the bank identify applicants with different levels of financial risk.

- **Low Risk:** Application can be approved.
- **Medium Risk:** Application can be sent for further review.
- **High Risk:** Application can be rejected or given stricter conditions.

This can help reduce potential loan defaults and improve the consistency of financial decisions.

13. Result

The Decision Tree successfully classified the given loan application as **Approved** based on the provided financial factors. The algorithm provides a simple and scalable approach for supporting loan approval and financial risk management.

14. Conclusion

The proposed Decision Tree-based system provides an efficient method for loan approval risk optimization. By considering credit score, income, debt-to-income ratio, and previous defaults, the bank can make systematic decisions while reducing default risk. The approach can also be extended to larger datasets and more financial features.